FULL-SYSTEM-BLUEPRINT — complete build document (Session 46, 2026-10-29)

> The **complete, self-contained** edition of the system plan: structure · architecture · workflows · blueprint · build instructions for every part — data gathering, databases, engines, non-LLM & LLM agents, trading workflows, research→backtest→alpha→paper→live, training/self-learning/self-upgrade, cell improvement, cross-cell testing & merging, swarm agents, comms, memory of every agent + memory management, chat with us, monitor/control, performance/accuracy/latency/error/bug/issue management — with pinpoint instructions to build, as pdf/md/html. Companion (executive overview): `FULL-SYSTEM-MASTER-PLAN.md`. Decisions trace to the four specialist docs and Sessions 1–45 in `planing conva/PLANNING-CONVERSATION-LOG.md`. Labels: ESTIMATE = projected/to-measure · MEASURED = on record · everything else = a locked decision from a source session.

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§1 · What we are building (one screen)

A private, from-scratch global intelligence ecosystem with a dedicated trading stack:

Row	Contains	Rule
Row 1 — public	Terminal · News portal · Media hub (LAST)	rights firewall; consumes engines only; never feeds back
Row 2 — foundation	Financial · News · World engines + shared machinery (registry, schemas, provenance, quality, bus, APIs, rights)	collect once, reuse many
Row 3 — private	**FinOS (trading stack — this plan's core)** · Cockpit (LLM agents, memory, approvals) · World Monitor (LAST)	intelligence never in fast paths

Invariants (non-negotiable): R1 collect-once · R2 rights at the boundary · R3 LLM never in hot paths · R4 PIT/survivorship discipline · R5 provenance everywhere · R6 versioned contracts · R7 train→eval→paper→approve→deploy · R8 HFT honesty · R9–R14 regulatory set · guards G1–G6 · kill cascade L0–L4 with out-of-band kill.

§2 · Complete structure

2.1 Monorepo layout (design target — created at W0a kickoff)

```
mani/
  rust/                 # hot-path kernel (Rust, edition 2024)
    rail/               # R1: validate · as-of · rights-tag · fan-out
    featurize/          # E1: in-process, pre-warmed state
    evaluate/           # E2: loads D5 artifacts, compiled formulas/ONNX
    govern/             # E5: deterministic rules
    exec/               # E6: scheduler, paper/shadow/live modes, adapters
    ledger/             # E7: append-only writer
  python/
    research/           # Mode R stack (PySR, MAPIE, arch, qlib as libs)
    gates/              # E4: CPCV · DSR · crisis replay · swarm harness
    seats/              # LLM seat runtimes (build/review/…/cockpit)
    tools/              # ops scripts, calendar, reconcile
  configs/              # subjects, schemas, rights classes, calendars
  compose/              # container stack (stores, bus, cockpit)
  vault/                # -> gfi Obsidian vault (docs, digests, decisions)
  graphify-out/         # code knowledge graph (auto, --update)
  docs/                 # the five specialist plans + this blueprint
```

Language boundary: Rust = R1/E1/E2/E5/E6/E7 (kernel); Python = research/gates/seats (edge). Contracts across the boundary (§3.3); no framework in the hot path.

2.2 Service inventory (Compose wraps; W0a)

Postgres (D3+D4, one instance two schemas) · QuestDB (D1) · NATS JetStream (D2) · rail process (R1) · cockpit chat server · Opensearch-free; plus build workers (Graphify `--update`, seat runs) — **rail is also able to run bare-metal outside Compose** for the tick path; Compose defines a cpu-pinned deployment (council fix A2).

§3 · Full architecture

3.1 Store ladder (every store, one job)

ID	Tech	Holds	Writers	Readers	Class
D0	Parquet NVMe + object offsite	years of ticks/bars/filings/news/world	R1 (+collectors in B1)	E4 replays, D6	bulk
D1	**QuestDB** (final; CH fallback)	live window days–weeks	R1 only	E1 pre-warm, dashboards, PIT replay	ms
D2	NATS JetStream	every live event, as-of stamped	R1	Mode T cells, seats	µs–ms fan-out
D3	Postgres	registry·schemas·rights·symbols·calendars·limits· `llm_tasks`	our services	everyone	ms
D4	Postgres **append-only**	fills·P&L·slippage·G4 trials·audit·CA ≥5 yr	E7, G4, seats-audit	research labels, ops	ms async
D5	fs + PG index	frozen artifacts (feature-hash+schema-signed), scores, citing-artifacts	gates+owner	E2 at boot	ms boot
D6	DuckDB	vectorized research scans	one process	Mode R	s–min
—	graphify-out/graph.json	code+docs knowledge graph	Graphify --update	seats (recall)	ms

Design rules: hot path NEVER queries a store per tick (rule 2) · research reads columnar only (rule 3) · D4/D5 append-only/signed at DB-role level (rule 4) · cold in D0, warm in D1, hot in RAM, truth in D3/D4/D5.

3.2 Engine stack (every job, one engine)

Engine	What	Where
E1	features: bars·ticks·Greeks·indicators·microstructure·pattern flags (Rust, pre-warmed)	hot
E2	frozen-artifact evaluator each tick (Rust+compiled)	hot
E3	training orchestrator (SB3/PyTorch wrapper, nightly offline, Colab T4 lane)	offline
E4	THE gate: E4-SIM swarm (10^4 worlds) · E4-BT vectorized+event-driven(fill model) · E4-WF walk-forward+CPCV · DSR≥0.95 · crisis replay	offline→
E5	governor: stops·size caps·news ladder·kill-switch·ban (zero ML)	hot
E6	execution: PAPER · SHADOW (zero order authority) · LIVE (adapters Upstox/cTrader/Hyperliquid; Nautilus under interface, swap-in)	hot
E7	ledger: fills vs expected·slippage·per-style cost curves→D4	async

Ladder (a candidate's life, same signed artifact at every rung, same E1 code — bit-for-bit): E4-SIM (minutes) → E4-BT vectorized (seconds) → E4-BT event (fills) → E4-WF CPCV+DSR → E6-PAPER (days) → E6-SHADOW (3–6 mo) → owner (≤5/week) → E6-LIVE → E7 → decay monitor → Mode R.

3.3 Adopt-vs-build + five interfaces

Build (no OSS substitute): R1 rail · E1 · E2 · E3 wrapper · E4 gate · E5 · E6 · E7+cost model+CA exporter · D5/G4 · PIT replay · rights firewall. **Adopt:** Parquet · QuestDB · NATS · Postgres · DuckDB · PySR · MAPIE · arch · sklearn/PyTorch/SB3 · NumPy/Numba · qlib (pin; W1) · hftbacktest (fill-model checklist; crypto-direct only) · Nautilus (LGPL fallback behind the interface) · + (from the 13 repos: bridge guard/approval/cron from brigade, swarm DAG from ruflo, personas, day-trader checks, quant-mind schema, Vibe anti-fabrication, timecopilot gate pattern, Graphify) — full map in `LLM-FLEET-BLUEPRINT.md` §3.1.

Contracts: 1 storage (rail-only writes) · 2 research (libs output candidates, never artifacts) · 3 artifact (D5 signs → E2 compiles) · 4 execution (E6 only order path) · 5 audit (everything → D4/G4).

§4 · Data gathering ("grathing") pipeline

Element	Plan
Priority	**data first → agents second → history grows third** (S42 owner order)
Tiers	T1 full tick+depth = tradable set (0.1–0.3 TB/yr ESTIMATE) · T2 world/news daily/1-min+reference (50–300 GB/yr) · T3 world pin-to-pin = per-research-need on demand, never blanket
Catalog	master TODO 1,114 rows / 18 volumes (S43) drives collectors; every row has license class + verify state
Registry	D3 source-registry seeded from the TODO: source_url, fetch time, parser version, rights class (R5/rule)
B1 gate	pull 6–10y of market+news+world → B-GATE: domain backfill verified + gaps table open before ANY agent trains
Blocked sources	rows keep verify/quote actions; explicitly marked in the gaps table, never silently skipped
Free/paid split	free verified bundle first; paid MVP rows collide at O12 (PIT fundamentals) and O15 (NSE membership)
Storage target	16 TB accepted (S43); D0 footprint 4–16 TB range; object offsite mandatory before 100s-of-GB backfills

§5 · Trading workflows (per market, locked fleet §2–§5)

Market	Venue	Works like	Exits/guards
Stocks ≤59 (login + optionable weight names)	Upstox	options **BUYING only**, same-day: E5 spread check → news ladder → 0DTE only long-gamma rule D4	rule ladder 1–6 + kill; no short gamma
FX Team 1 (7 pairs + 4 commodities)	cTrader ECN	3 styles/pair (session·carry·news), regime router picks ≤1	cascade-flatten · weekend-gap guard · funding-window print · kill
FX Team 2 (XAU/USD + WTI)	cTrader, **separate account**	pure scalping push-mode (requires tick+depth microstructure food = E1 block)	session-closed flat; every-tick stop
Crypto (BTC/ETH/HYPE)	Hyperliquid perps	4 styles: funding · liquidation · weekend · PAIR stat-arb	funding-window · 5-liquidity-gate exit (no bid → underlying hedge close-only) · kill

Common: E2 evaluates frozen artifact each tick → E5 governs size/stops → E6 routes (validated, idempotent client IDs) → fill async to E7 ledger → D4 · reconcile-before- orders on restart (invariant) · full replay of any decision on demand.

§6 · Research → alpha workflow (gaps & alpha finding — the machine that finds trades)

1. **COLLECT** (R1-stamped) → **LABEL** triple-barrier, net-of-costs, from the **signal log** (all signals, taken or not — not just fills).
2. **FIND ALPHA** in Mode R (offline swarm): GP formula search (PySR) · ML discovery (E3) · meta-labeling (López de Prado pattern; single-source evidence — adopt cautiously C14) · factor/parameter sweeps → screened by E4-SIM (10^4 worlds).
3. **GAP FINDING 1 (research gaps):** G4 coverage maps = swarm survival fractions show untested parameter regions → the next search batch targets those cells.
4. **GAP FINDING 2 (data gaps):** fleet §9.1 register + coverage per symbol/time → collector tasks, in-DB gap calendar; untested data shapes are tickets, not guesses.
5. **GATE:** E4-BT (vectorized → event-driven with fill model per hftbacktest checklist) → E4-WF CPCV (purge = label horizon + embargo; folds day-grouped across symbols; untouched final holdout) → **DSR ≥ 0.95 vs decorrelated trials (G4)** → crisis replay (EOD index-option history) → net-of-cost. Multiple-testing honesty: 10^3 candidates/style cap; failure = scope does not trade.
6. **PAPER** (E6-PAPER, real path sim fills) → **SHADOW** (3–6 mo, zero order authority) → **owner approves ≤5/week** → D5 signed → live on next boot.
7. **MONITOR:** rolling IC · PSI/edge decay (net-IC vs frozen) → retrain/retire.

Scores from LLM seats ride this ladder identically: no score is trusted before the **E2 edge-lift gate** shows it lifts measured edge in replay repeatedly (S44/§4).

§7 · Training · self-learning · self-upgrade · cell improvement · cross-cell

Mechanism	How
Training	E3 nightly on Colab T4 (verified lane) or local; every run G4-numbered; nothing enters Mode T without §6 gates
Self-learning	weekly replay-test batch: every still-trusted artifact re-scored on last windows; decay list → research tickets; retraining triggered by **edge decay**, never raw PSI (C9)
Self-upgrade (ceiling)	"upgrade" = new frozen artifacts through the full gate. NEVER: live code rewrite (G1), order placement from research (G3), LLM in R1/R2 (R3), style enable without E4 (R7/R-full r7)
Cell improvement	shadow feedback + E7 slippage curves tune exits/cost model via normal PRs (LLM build seat) — same gates, no live bypass
Cross-cell testing	monthly: overlapping artifacts (same family, many cells) → union-instrument test set → re-clear FULL E4 on the union
Cross-cell merging	on pass: merged artifact replaces **per-cell** retires (owner-approved); G4 logs lineage; rollback = D5 prior version
Swarm testing	cross-style/cross-market swarm sims in E4-SIM (correlation, exposure, drawdown interaction) before any merge lands

§8 · Swarm agents (what "swarm" means here — both kinds)

Kind	What	Powered by
Fleet swarm	10^4 deterministic simulated worlds for any strategy/question (vectorized NumPy/Numba) — E4-SIM	our engine, seed+param-hash replay
LLM crew swarm	multi-seat work on one mission (research envelope → analysis → scoring → ops) with DAG deps	ruflo `@claude-flow/swarm` task-orchestrator contract (ported) · Vibe-Trading preset patterns (tool/skill whitelists, anti-fabrication) · agency-agents personas · our NATS+Postgres queue
Cross-cell swarms	any new style tested against existing cells' exposure in sim before merge	§7 swarm testing

Crew config schema: `crew.yaml` (roster of seat ids · depends_on DAG · tool whitelist · out_artifact schema · budget cap · approval level). CI lint enforces roster/administrative fields (agency-agents runbook contract pattern).

§9 · Communication & wiring (the whole wire)

Five transports only: 1) NATS subjects — `market.*` (R1) · `signal.*/order.*/ risk.*/ledger.*` (R2) · `seat.*` (LLM); queue-groups = load-balance, durable consumers = independent replay, partitioned per-key ordering. 2) Files — D0/D5, single-writer. 3) SQL — D1/D3/D4 concurrent servers (headroom = G1 bench). 4) Embedded DuckDB — single D6 process, never critical path. 5) In-proc calls — E1/E4 workers, µs.

Latency budgets (ESTIMATE until gates measure):

Hop	Budget
E1/E2 decision (pre-warmed)	<1 ms; µs-class cell internal
R1→D2 fan-out → cell receive	≤ ~2 ms P99 (W0b measure)
E1→E5→E6→wire (order)	<5 ms internal; **venue RTT dominates** (Upstox ~45 ms vendor-claim · Hyperliquid ~600–700 ms public API · cTrader GA-routed)
D1 reads	boot/replay/dashboards only — never per-tick
seat tasks	seconds–minutes, intentionally async
research scans	seconds–minutes off-peak

Safety comms: out-of-band watchdog process (no app dependency) · venue-side max-loss order · one-action mobile kill · heartbeat to off-box.

§10 · Every agent's memory + memory management

10.1 Non-LLM fleet (75 brains)

Type	Where	Writer
working	cell process state (tick/feature/position/DD state)	the cell itself
episodic	**E7 ledger** + **G4 trial log** + shadow history	E7/G4
semantic	**R3 registry**: frozen artifacts, cost model, calendars, limits, ban status, regime→style map	gates+owner only
procedural	the code (formulas, exits ladder, governor rules, runbook)	git PR only

10.2 LLM seats

Type	Where	Writer
working	task lease payload (`llm_tasks`)	seat, on lease
episodic	D4 audit rows (every run: tokens, result, artifacts)	seat runtime
semantic	D5 artifacts + graph.json + vault notes	versioned commits only
procedural	skill files	PR + review
owner rules	vault note	cockpit seat only

10.3 Management policies

Write authority is **enforced, not prose** (DB roles; no seat can UPDATE D4; no seat can write D5 except gated artifact path). Retention: D4 ≥5 yr; G4 forever; `llm_tasks` trim after N days (digest summarization first); vault is curated by hand/cockpit. Recall stack: Graphify graph for code/architecture; vault digest notes for decisions; substrate (brigade Tideline patterns vs ruflo memory) decided by **W0c toy-shootout** (measured). Obsidian = declarative decision memory already living in `gfi/`; formal note types: decision · incident · runbook · research-digest · training-record.

§11 · Chat with us + full control/monitor panel (cockpit)

Chat interface (owner ⇄ system):

- One keyboard surface (opencode/claude session as seat-12) + one web panel (OpenStock- pattern UX ideas re-implemented; session-clock, alert schema, cache pools as concepts).
- Every reply labeled ESTIMATE/MEASURED; every quote carries its artifact id.
- Owner actions by tier: L0 chat answers · L1 sandbox checks · **L2 sign-offs** (news ladder up/down, budget raise, ticket triage) · **L3 live-decisions** (promotions, kill re-arm, deploy) — approval cards in chat; the ≤5/week promotion quota lives here.
- Everything chat says/does streams to D4 + weekly digest → vault.

Monitor surfaces (G2): | Surface | Data | |---|---| | Cell health | per-cell heartbeat, state (E1/E2/E5), last decision id, kill-status | | Portfolio | D4-derived exposure/net across the 3 sub-accounts | | Market state | session clock per venue, kill-state, ban status | | Data | gap calendar, D0/D1 usage vs 16-TB, collector backlog | | LLM layer | token burn/seat/day (G9), eval-harness replays, failed tasks | | Research | G4 trial counts, DSR status of trusted set, coverage map heat | | Ops | incident queue, restart/restore drill results, backup freshness |

Execution/kill: cockpit never sits in the order path; the L0–L4 cascade and the out-of-band watchdog do. Cockpit *asks* E5/E6 orbitally (commands to gate; never direct).

§12 · Performance, accuracy, latency, error, bug & issue management

Discipline	Mechanism
Latency	per-hop budgets (§9) measured at gates (W0a/W0b) into D3 dashboards; venue RTT reported with vendor-claim labels
Accuracy	replay determinism bit-for-bit; eval-harness stubs (timecopilot FunctionModel pattern) test seat flows without any provider; accepted-model stiction via drift triggers
Errors/bugs	ticket taxonomy (bug · incident · data-gap · drift · ops) → build seat reproduces with **flight-recorder** (E1/E2/E5/E7 traces replay to same decision id) → fix+regression test → gates re-run
Error budget	per-lane error budget (order path, rail, seat runtime) — p95/p99 targets tracked; budget breach = auto shadow-mode for that lane (ESTIMATE, tune at W0c)
Issue mgmt	cockpit queue surfaces open tickets with owner-assignable priority; weekly digest includes burn-down
Chaos	10×-restart · feed-stale · venue-down · kill-switch drills monthly (runbook §6A)
Code gates	`cargo clippy -- -D warnings` + `cargo test` · `tsc --noEmit` (TS seats) · `uv run pytest` (Python) · review + cybersec seats · secret scan pre-commit

§13 · Build order with pin-point instructions (waves + gates)

Wave	Build	Exit pins (must be real command output)
W0a	monorepo skeleton §2.1 · Compose stack · R1 rail · D0/D1 pick (5 gates) · PIT replay · G1 concurrency bench	replay identical ×2 · D1 gates file · bench JSON
B1	collectors market/news/world · 6–10y fill per tiers · D3 gaps table	B-GATE: rows verified · gaps open-but-justified · no training before
W0b	D4/D5 roles+signing · E1/E2/E5/E7 skeletons · L0–L4 + watchdog · IPC fan-out bench	restart ×10 · fan-out p99 · kill drill log
W0c	LLM control plane: `llm_tasks` · seat runtimes on brigade-guards/ruflo-DAG port · approval bridge L0–L3 · Graphify+Obsidian fold-in · G1 vault · G2/G9 dashboards · substrate shootout · cockpit v1 (chat only)	toy-build cold start · 10× Compose restart · Graphify delta measured · shootout numbers
W1	D6 research cache · PySR/MAPIE/arch wiring · E4-SIM/BT/WF · E6-PAPER · first full style through §6 to D5	pipeline run log · D5 signed artifact
W2	cost model verified (O8) · cTrader adapter · proton JetStream-MV trial (own gates) · seats news/data-QA/compliance live · reconcile E7 vs broker	paper fills + ledger match
W3	first cells shadow · NY4 lane planning · Live requires owner L3 + shadow history	shadow P&L/IC files
W4–W5	tick+depth feeds → E1 microstructure · Hyperliquid node option · crypto cells · ML lane (E3) live	microstructure replay · HL colo decision
Stage 6	cockpit GA (panel + chat approvals; O16 resolved)	L3 flow demo · weekly digest in vault
Stages 7–10	Terminal/Portal/Media (rights firewall) · World Monitor LAST · HFT subsystem only if measured	Row-1 replica isolated · no hot-path impact test

Week-1 instructions (concrete): create monorepo §2.1 skeleton · bring up Compose stack with Postgres/QuestDB/NATS + healthchecks · write R1 rail minimal loop (validate → stamp → publish to `market.*` + dump D0 parquet) · replay harness CLI · G1 bench harness stub · `docs/` import (5 plans + this one) · vault note types created · Graphify baseline graph build (`graphify` installed at W0a; `--update` mode understood) · open the W0a gate document to record every measurement as it happens.

§14 · Risk register (unchanged standings)

single-box fragility (offsite + compose snapshots before big backfills) · venue RTT domination · license drift re-verify cadence · India data gaps pre-2022 (O12/O15) · single-source research evidence (C14) · new-code failure modes of the cockpit layer (reviewed at trading-code parity) · LLM seat cost/perf oscillation → budget caps + EN plugboard G9.

§15 · Open decisions (status)

Carried from log with owners/waves: O3 F&O granularity · O4 off-index names · O5 first style · O6 FX scalping add · O7 hedge executor · O8 cTrader prod limits · O9 Upstox algo-tag · O10 order-balance · O11 HL FIU-IND/FEMA/tax · O12 PIT fundamentals · O13 ToS checklist · **O14 model routing/privacy (W0c)** · **O15 NSE membership vendor (W2)** · **O16 cockpit runtime (W0c)** · ~~O1 resolved (S43)~~.

§16 · Sources

All content synthesizes locked decisions from: FULL-SYSTEM-PLAN.md · AGENT-FLEET-BLUEPRINT.md · ENGINE-DATABASE-PLAN.md (incl. §19 TSDB final) · LLM-FLEET-BLUEPRINT.md · plan log S1–S45. No new external claims; every prior vendor or author-reported number remains labeled as in its source doc.

Session 46 log entry + commit hash recorded in the planning log.